2D Arrays Master Practice Sheet

1. Compulsory Questions (Must Do Basics)

- Print 2D Array (Easy)

Take input for a 2D array and print it in matrix form.

- Row-wise Sum (Easy)

Calculate and print the sum of elements for each row in a 2D array.

- Column-wise Sum (Easy)

Calculate and print the sum of elements for each column in a 2D array.

2. Best Logic Building Questions

- Transpose of a Matrix (Medium)

Convert rows to columns and columns to rows in a square matrix.

- Diagonal Sum (Medium)

Find the sum of the main diagonal and secondary diagonal elements.

- Check Identity Matrix (Medium)

Check if a square matrix is an identity matrix (1s on the main diagonal, 0s elsewhere).

3. Hard Level Questions

- Spiral Print (Hard)

Print the elements of the 2D array in spiral order (clockwise).

- Wave Print (Hard)

Print the elements of the 2D array in wave form (column-wise zigzag).

- Matrix Multiplication (Hard)

Multiply two matrices if dimensions are compatible.

4. More Hard Questions

- Rotate Matrix by 90 Degrees (Very Hard)

Rotate a square matrix 90 degrees clockwise in-place.

- Search in Sorted Matrix (Very Hard)

Efficiently search for an element in a row and column-wise sorted 2D array.

- Set Matrix Zeroes (Very Hard)

If any cell is 0, set its entire row and column to 0.

5. Extra Practice (Optional Fun Challenges)

- Checkerboard Pattern (Medium)

Create and print a checkerboard pattern (like a chessboard) using 1s and 0s.

- Sum of Boundary Elements (Medium)

Find the sum of all the boundary (outermost) elements of the matrix.

- Snake Pattern Print (Hard)

Print the matrix in a snake-like pattern (left-to-right on even rows, right-to-left on odd rows).